

Neural Surrogate Refined with SAT3

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What was the previous V2 Neural Surrogate doing

A completely black-box model

Use input vectors: $k_y, \epsilon, \hat{S}, q, \omega_{Ti}, \omega_{Te}, \omega_n$

Predicting: $\gamma(k_y), \omega(k_y)$

Details available at:

<https://github.com/Jekyll-Chan/Neural-Surrogate-TatyanaV2.git>

The upgraded mapping

Input Vectorization

Local Equilibrium Parameters : $k_y, \epsilon, \hat{S}, q, \omega_{Ti}, \omega_{Te}, \omega_n$

Isotope Scaling Input (Roadmap): **Ion Mass (A)** or rho_unit

To activate the ρ_{unit} factor in *SAT3* for capturing the anti-gyroBohm effect

The TEM saturation level explicitly depends on the unit gyro-radius $\rho_{unit} \propto \sqrt{A}$

Neural Surrogate Inference (The Linear Core)

Acts as the high-speed engine to predict linear instability characteristics

Process : species-resolved linear targets : $\gamma_e, \gamma_i, \omega_e, \omega_i$

Successfully implemented with a $\sim 0.9\%$ Median Relative Error. These species-resolved outputs serve as proxies for the **linear energy weights** W_L required for final flux integration.

Conversion Layer (Analytical Feature Extraction)

Before applying nonlinear rules, the model must extract key physical "landmarks" from the predicted linear spectra.

Uses an analytical grid scan to identify k_{max} and γ_{max}

Peak Drive Identification

k_{max} is defined as the wavenumber where the linear drive density γ/k_y is maximized

Calculates the mode ratio to diagnose the dominant instability

Mode Diagnostics

This allows the model to switch between ITG and TEM regimes with 100% threshold accuracy

SAT3 Algebraic Engine (The Nonlinear Saturation Rules)

The extracted linear features are fed into the SAT3 formulas to determine the saturated state.

ITG Scaling: Saturated potential $\propto \gamma_{max}^2 / k_{max}^5$

TEM Scaling: Saturated potential $\propto \gamma_{max}^2 \rho_{unit} / k_{max}^4$

Mode-Dependent Scaling: The model applies specific scaling laws based on the diagnosed regime

Spectral Shape Constraint: The model is designed to predict the radial width σ_{ky} and enforce a robust -2.42 power-law relationship to synthesize the full 1D potential spectrum

Final Transport Output (Synthesized Fluxes)

Integrates the saturated potential spectrum with linear phase information to produce the transport results.

Flux Summation:

The total radial heat flux Q_s is synthesized by Integrating the species-resolved linear weights and the SAT3-predicted saturated potentials

Outcome:

Accurate, millisecond-scale prediction of nonlinear heat and particle fluxes that maintain physical consistency across different hydrogen isotopes

V3 Neural Surrogate

Note: Simplification vs. Dudding's R_{ei} , justified by the clean separation shown in the ω distribution plot
Mode diagnosed via $\text{sign}(\omega)$: $\omega > 0 \rightarrow \text{ITG}$, $\omega < 0 \rightarrow \text{TEM}$, consistent with GENE/TGLF convention

Mathematical principles behind

Full Nonlinear gyrokinetic equation:

$$\frac{\partial h_s}{\partial t} + v_{\parallel} \hat{b} \cdot \nabla h_s + \mathbf{v}_{E \times B} \cdot \nabla h_s = \frac{e_s F_{M,s}}{T_s} \frac{\partial \langle \phi \rangle_s}{\partial t} + C[h_s]$$

That is:

$$\frac{\partial h_s}{\partial t} + (v_{\parallel} \hat{b} + \mathbf{v}_{d,s}) \cdot \nabla h_s + \frac{c}{B} \{ \langle \phi \rangle_s, h_s \} = \frac{e_s F_{M,s}}{T_s} \frac{\partial \langle \phi \rangle_s}{\partial t} + C[h_s]$$

Where:

$(v_{\parallel} \hat{b} + \mathbf{v}_{d,s}) \cdot \nabla h_s$: free-streaming + magnetic drift

$\frac{c}{B} \{ \langle \phi \rangle_s, h_s \}$: nonlinear $E \times B$ advection

$\frac{e_s F_{M,s}}{T_s} \frac{\partial \langle \phi \rangle_s}{\partial t}$: adiabatic drive

$$\frac{\partial h_s}{\partial t} + (v_{\parallel} \hat{b} + \mathbf{v}_{d,s}) \cdot \nabla h_s + \frac{c}{B} \{ \langle \phi \rangle_s, h_s \} = \frac{e_s F_{M,s}}{T_s} \frac{\partial \langle \phi \rangle_s}{\partial t} + C[h_s]$$

Where:

$\langle \phi \rangle_s = J_0(k_{\perp} \rho_s) \phi$ is the gyro-averaged potential (*Bessel-function ring average*)

$F_{M,s}$ is the Maxwellian background

$\{\cdots\}$ is the Poisson bracket in (x, y)

The magnetic drift velocity is:

$$\mathbf{v}_{d,s} = \frac{v_{\parallel}^2 + v_{\perp}^2/2}{\Omega_s} \hat{b} \times \nabla \ln B$$

Closure is provided by quasi-neutrality:

$$\sum_s \frac{e_s^2 n_s}{T_s} [\phi - \langle \phi \rangle_s] = \sum_s e_s \int J_0(k_{\perp} \rho_s) h_s d^3 v$$

Linearisation → TGLF eigenvalue problem

Fourier-decompose all fluctuating quantities in the field-aligned coordinates (x, y, θ) :

$$h_s = \sum_{k_x, k_y} \hat{h}_{s, k_z, k_y}(\theta) e^{i(k_z x + k_y y - \omega t)}$$

Dropping the nonlinear bracket $\{\langle \phi \rangle_s, h_s\}$ and substituting:

$$(-i\omega + v_{\parallel} \partial_{\theta} + \omega_{d,s}) \hat{h}_{s,k} = \frac{e_s F_{M,s}}{T_s} (-i\omega + i\omega_{*s}^T) J_0(k_{\perp} \rho_s) \hat{\phi}_k$$

The diamagnetic drift frequency with temperature gradient is:

$$\omega_{*s}^T = \omega_{*s} \left[1 + \eta_s \left(\frac{v^2}{v_{ts}^2} - \frac{3}{2} \right) \right], \quad \omega_{*s} = \frac{k_y T_s}{e_s B_{\text{unit}}} \cdot \frac{a}{L_{ns}}$$

Where $a/L_{n_s} = -a(\partial \ln n_s / \partial r)$ is just the TGLF input *omn*.

Similarly a/L_{T_s} maps to *omt_i* and *omt_e*.

TGLF's fast eigenvalue solver gives complex $\omega = \omega_r + i\gamma$ for each (k_x, k_y) — that's what V2 surrogates.

Turbulent flux in gyrokinetics

Once saturated, the radial heat flux for species S is an *ensemble* + *flux-surface* average:

$$Q_s = \left\langle \nabla x \cdot \frac{\hat{b} \times \nabla \delta \phi}{B} \delta p_s \right\rangle_{\text{Ens, FS}}$$


Inserting Fourier representations and using the reality condition $\delta \hat{f}_{-k_x, -k_y} = \delta \hat{f}_{k_x, k_y}^*$, this reduces to a sum over positive k_y only (derivation in Dudding Appendix B):

$$Q_s = \sum_{k_y > 0} 2 \sum_{k_z} \frac{k_y \text{Im} [Z_{s, k_x, k_y}^*]}{B_{\text{unit}}} \left\langle |\delta \hat{\phi}_{k_z, k_y}|^2 \right\rangle_{\theta, t}$$

where

$Z_{s, k_x, k_y} \equiv \delta \hat{p}_{s, k_x, k_y} / \delta \hat{\phi}_{k_x, k_y}$ encodes the pressure-potential phase relationship — the key physics of whether turbulence drives transport.

Quasilinear weight and QLA decomposition

Define the linear weight for each k_y as a k_x -weighted average:

$$W_{s,k_y}^L = \left\langle \frac{k_y \text{Im} [Z_{s,k_x,k_y}^{*L}]}{B_{\text{unit}}} \right\rangle = \frac{\sum_{k_x} \frac{k_y \text{Im} [Z_{s,k_x,k_y}^{*L}]}{B_{\text{unit}}} |\delta \hat{\phi}_{k_x,k_y}|_{\theta,t}^2}{\sum_{k_x} |\delta \hat{\phi}_{k_x,k_y}|_{\theta,t}^2}$$

The quasilinear approximation (QLA) function Λ_{s,k_y} measures how well the linear phase approximates the nonlinear one:

$$\Lambda_{s,k_y} \equiv \frac{W_{s,k_y}^{NL}}{W_{s,k_y}^L}$$

Quasilinear weight and QLA decomposition

If the QLA were exact, $\Lambda_{s,k_y} = 1$ everywhere. SAT3 fits mode-dependent constants:

$$(\Lambda_{ITG}^Q = 0.75, \Lambda_{TEM}^Q = 0.6, \Lambda_{ITG}^R = 1.1, \Lambda_{TEM}^R = 0.6)$$

The flux becomes

$$Q_s = 2 \sum_{k_y > 0} \Lambda_{s,k_y} W_{s,k_y}^L \left[\frac{\langle |\delta \hat{\phi}_{k_y}|^2 \rangle_{x,\theta,t}}{\Delta k_y} \right] \Delta k_y$$

And $\left[\frac{\langle |\delta \hat{\phi}_{k_y}|^2 \rangle_{x,\theta,t}}{\Delta k_y} \right]$ is exactly the SATURATION RULE OUTPUT! (Cheers~)

This factorisation is central: the linear solver supplies W_{s,k_y}^L and Λ_{s,k_y} ;

The saturation rule supplies the grid-independent 1D potential spectrum.

SAT3 — 2D spectrum model

The 2D spectrum $\langle |\delta \hat{\phi}_{k_x, k_y}|^2 \rangle_{\theta, t}$ in k_x is observed to be an approximately even, single-peaked function.

Taylor-expanding its inverse:

$$\frac{1}{\langle |\delta \hat{\phi}_{k, k_y}|^2 \rangle_{\theta, t}} = C_0(k_y) + C_1(k_y)(k_x - K)^2 + C_2(k_y)(k_x - K)^4 + \dots$$

Inverting and solving for the first three k_x -moments analytically :

(Dudding Appendix E, via partial fractions on the integral $\int_{-\infty}^{\infty} (E + Fu^2 + u^4)^{-1} du$)

$$\langle |\hat{\phi}_{k_x, k_y}|^2 \rangle_{\theta, t} = \frac{\langle |\hat{\phi}_{k_x = \langle k_x \rangle, k_y}|^2 \rangle_{\theta, t}}{1 + \left(\left(\frac{\pi \langle |\hat{\phi}_{k_x = \langle k_x \rangle, k_y}|^2 \rangle_{\theta, t} \sigma_{k_y}}{\Delta k_x \langle |\hat{\phi}_{k_y}|^2 \rangle_{x, \theta, t}} \right)^2 - 2 \right) \left(\frac{k_x - \langle k_x \rangle}{\sigma_{k_y}} \right)^2 + \left(\frac{k_x - \langle k_x \rangle}{\sigma_{k_y}} \right)^4}$$

$$\left\langle \left| \hat{\phi}_{k_x, k_y} \right|^2 \right\rangle_{\theta, t} = \frac{\left\langle \left| \hat{\phi}_{k_x = \langle k_x \rangle, k_y} \right|^2 \right\rangle_{\theta, t}}{1 + \left(\left(\frac{\pi \left\langle \left| \hat{\phi}_{k_x = \langle k_x \rangle, k_y} \right|^2 \right\rangle_{\theta, t} \sigma_{k_y}}{\Delta k_x \left\langle \left| \hat{\phi}_{k_y} \right|^2 \right\rangle_{x, \theta, t}} \right)^2 - 2 \right) \left(\frac{k_x - \langle k_x \rangle}{\sigma_{k_y}} \right)^2 + \left(\frac{k_x - \langle k_x \rangle}{\sigma_{k_y}} \right)^4}$$

Where $\sigma_{k_y} = \sqrt{\langle k_x^2 \rangle - \langle k_x \rangle^2}$ is the radial spectral width — the central new degree of freedom in SAT3.

This generalises SAT1/2's squared-Lorentzian assumption ($\sigma_{k_y}^2 C_1 = 2$ fixed) by retaining it as a free, k_y -dependent quantity.

SAT3 — core relation

Summing the 2D spectrum over k_x via Parseval's theorem gives the 1D potential:
(The key empirical discovery of Dudding et al. is that across all 43 CGYRO simulations)

$$\frac{\langle |\delta \hat{\phi}_{k_y}|^2 \rangle_{x,\theta,t}}{\Delta k_y} = \sigma_{k_y}^{c_1}, \quad c_1 \approx -2.42 *$$

This is a power-law relation between the zeroth moment (1D potential) and the second moment (radial width) of the 2D spectrum — a *conserved-like* quantity in electrostatic turbulence with no prior theoretical explanation.

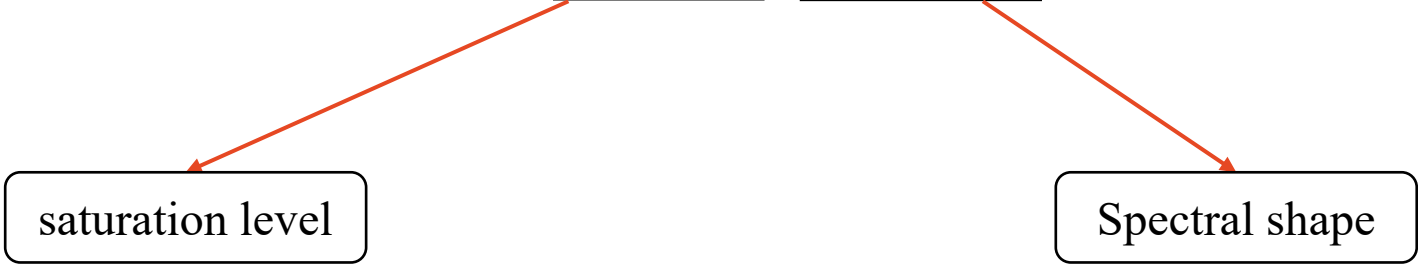
Physically it says: a broader radial spectrum implies a smaller amplitude, independent of the peak value.

c_1 is universal across all cases;

c_0 is case-dependent and absorbs the saturation level.

Spectral shape model

Evaluating on the previous slide at a reference point k_0 and dividing, the 1D potential becomes a product of a saturation level and a spectral shape function:

$$\frac{\langle |\delta \hat{\phi}_{k_y}|^2 \rangle}{\Delta k_y} = \boxed{\frac{\langle |\delta \hat{\phi}_{k_0}|^2 \rangle}{\Delta k_y}} \cdot \boxed{\left(\frac{\sigma_{k_y}}{\sigma_{k_0}} \right)^{-2.42}}$$


The diagram illustrates the decomposition of the equation above. Two red arrows originate from the boxed terms in the equation. The first arrow points from the first boxed term, $\frac{\langle |\delta \hat{\phi}_{k_0}|^2 \rangle}{\Delta k_y}$, to a rounded rectangular box labeled "saturation level". The second arrow points from the second boxed term, $\left(\frac{\sigma_{k_y}}{\sigma_{k_0}} \right)^{-2.42}$, to a rounded rectangular box labeled "Spectral shape".

saturation level

Spectral shape

The radial spectral width σ_{k_y} is modelled as a piecewise quadratic (eq. 20), reflecting the observed minimum at k_{min} corresponding to the peak of the potential spectrum:

$$\sigma_{k_y} = \begin{cases} ak_y^2 + bk_y + c & 0 < k_y \leq k_p \\ (2ak_p + b)k_y + c - ak_p^2 & k_y > k_p \end{cases}$$

$$\sigma_{k_y} = \begin{cases} ak_y^2 + bk_y + c & 0 < k_y \leq k_P \\ (2ak_P + b)k_y + c - ak_P^2 & k_y > k_P \end{cases}$$

with $k_P = 2k_{min}$ ensuring C^1 continuity.

Since $d\sigma_{k_y}/dk_y|_{k_{min}} = 0$ gives $a/b = -1/(2k_{min})$, the shape can normalise to:

$$\frac{\sigma_{k_y}}{\sigma_{k_0}} = \frac{-\frac{2k_y^2}{2k_{min}} + k_y + \frac{c}{b}}{-\frac{2k_0^2}{2k_{min}} + k_0 + \frac{c}{b}}$$

with fitted parameters:

$$\begin{aligned} k_{min} &= 0.685 k_{max} , \\ c/b &= -0.751 k_{max} , \\ k_0 &= 0.6 k_{min} \end{aligned}$$

Saturation level — ITG vs TEM

The characteristic linear scales k_{max} and γ_{max} are defined by:

$$\frac{d}{dk_y} \left(\frac{\gamma_{k_y}}{k_y} \right) \Big|_{k_{max}} = 0, \quad \gamma_{max} = \gamma_{k_y=k_{max}}$$

This is the wavenumber where linear growth-to-damping ratio (drive vs zonal mixing $k_y V_{ZF}$) is maximised.
The saturation level depends on mode type:

$$\frac{\langle |\delta \hat{\phi}_{k_0}|^2 \rangle}{\Delta k_y} = M \left(\frac{W_e^L}{\sum_i W_i^L} \Big|_{k_{max}} ; 0.8, 1.0, 3.3 \frac{\gamma_{max}^2}{k_{max}^5}, 12.7 \frac{\gamma_{max}^2 \rho_{unit}}{k_{max}^4} \right) B_{unit}^2$$

where $M(x; x_1; x_2; y_1 y_2)$ is a linear transition function.

The extra factor of $k_{max} \rho_{unit} \propto \sqrt{A}^{-1}$ in the TEM branch is the direct source of isotope scaling improvement: in TEM-dominated turbulence, $k_{max} \rho_{unit}$ has a slightly more positive $\alpha_A \approx -0.3$ dependence (rather than exact $A^{-0.5}$), (which gives the anti-gyroBohm scaling that SAT1/2 missed entirely).

$$\frac{\langle |\delta \hat{\phi}_{k_0}|^2 \rangle}{\Delta k_y} = M \left(\frac{W_e^L}{\sum_i W_i^L} \Big|_{k_{\max}} ; 0.8, 1.0, 3.3 \frac{\gamma_{\max}^2}{k_{\max}^5}, 12.7 \frac{\gamma_{\max}^2 \rho_{\text{unit}}}{k_{\max}^4} \right) B_{\text{unit}}^2$$

The ITG dimensional form

$$\gamma_{\max}^2 / k_{\max}^5$$

matches the balance between linear growth $\sim \gamma_{\max}$ and turbulent $E \times B$ advection

$$\sim k_{\max}^2 V_{E \times B} \sim k_{\max}^2 \cdot k_{\max} \phi / B \sim k_{\max}^3 \phi$$

giving

$$\phi \sim \gamma_{\max} / k_{\max}^3$$

and

$$\phi^2 / k_{\max}^2 \sim \gamma_{\max}^2 / k_{\max}^5$$

Full SAT3 flux

Assembling all components:

$$Q_s = 2 \sum_{k_y > 0} \Lambda_{s,k_y} W_{s,k_y}^L F_{s,k_y}(\gamma_E) \cdot M(R_{ei}; \dots) B_{\text{unit}}^2 \left(\frac{\sigma_{k_y}}{\sigma_{k_0}} \right)^{-2.42} \Delta k_y$$

where $R_{ei} \equiv |W_e^L / \sum_i W_i^L|_{k_{max}}$ is the electron-to-ion linear weight ratio, the $E \times B$ shear factor is (from SAT2):

$$F_{s,k_y}(\gamma_E) = \frac{\bar{W}_{s,k_y,k_x=k_{x0}}^L / \bar{W}_{s,k_y,k_x=0}^L}{\left(1 + \left(\alpha_x \frac{k_{x0}}{k_x^{\text{model}}} \right)^{\sigma_x} \right)^2 \left(1 + \left(\frac{k_{x0}}{k_x^{\text{model}}} \right)^2 \right)^2}$$

and the QLA functions use the mode transition:

$$\Lambda_{s,k_q}^Q = M(R_{ei}; 0.8, 1.0, 0.75, 0.6), \quad \Lambda_{s,k_y}^I = M(R_{ei}; 0.8, 1.0, 1.1, 0.6)$$

Connecting to Surrogate V3

Now the architecture is clear 🤖 From the factorised flux equation, SAT3 requires exactly three scalar quantities computed from the full linear spectrum $\{\gamma_{k_y}, \omega_{k_y}, W_{s,k_y}^L\}$:

$$\mathcal{S} = \left\{ k_{\max}, \gamma_{\max} = \gamma_{k_y=k_{\max}}, R_{ei} = \left| \frac{W_e^L}{\sum_i W_i^L} \right|_{k_{\max}} \right\}$$

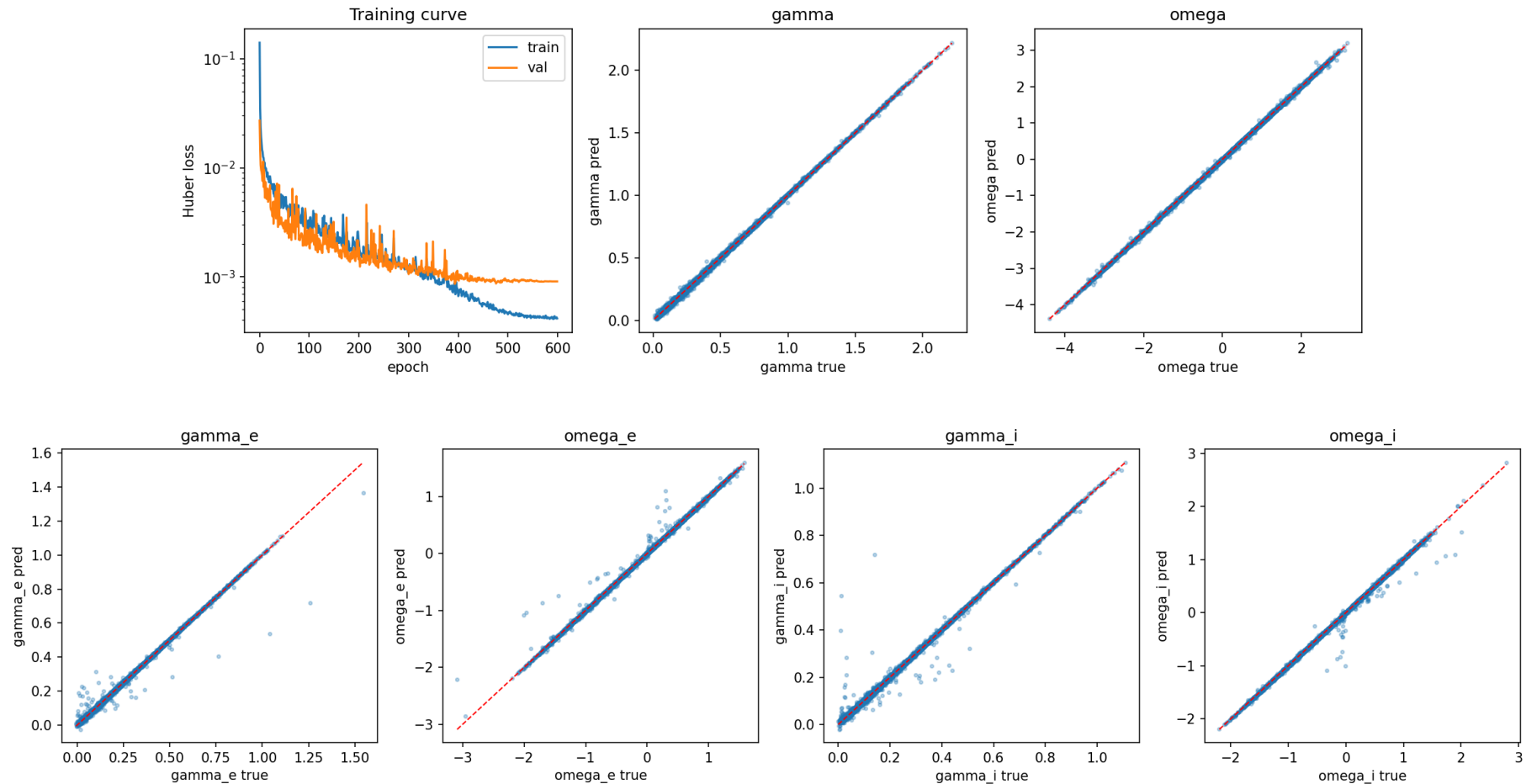
Once $\mathcal{S}$ is known, the entire k_y -integral is computable analytically. The Tatyana V2 outputs (γ, ω) at a single $(k_y, \mathbf{x}_{eq})$ can provide W_{s,k_y}^L at that point via:

$$W_{s,k_y}^L = \frac{k_y \operatorname{Im} [Z_{s,k_y}^{*L}]}{B_{\text{unit}}} \approx \left. \frac{k_y \operatorname{Im} [Z_{s,k_y}^{*L}]}{B_{\text{unit}}} \right|_{k_y=k_{\max}}$$

V3 therefore targets:

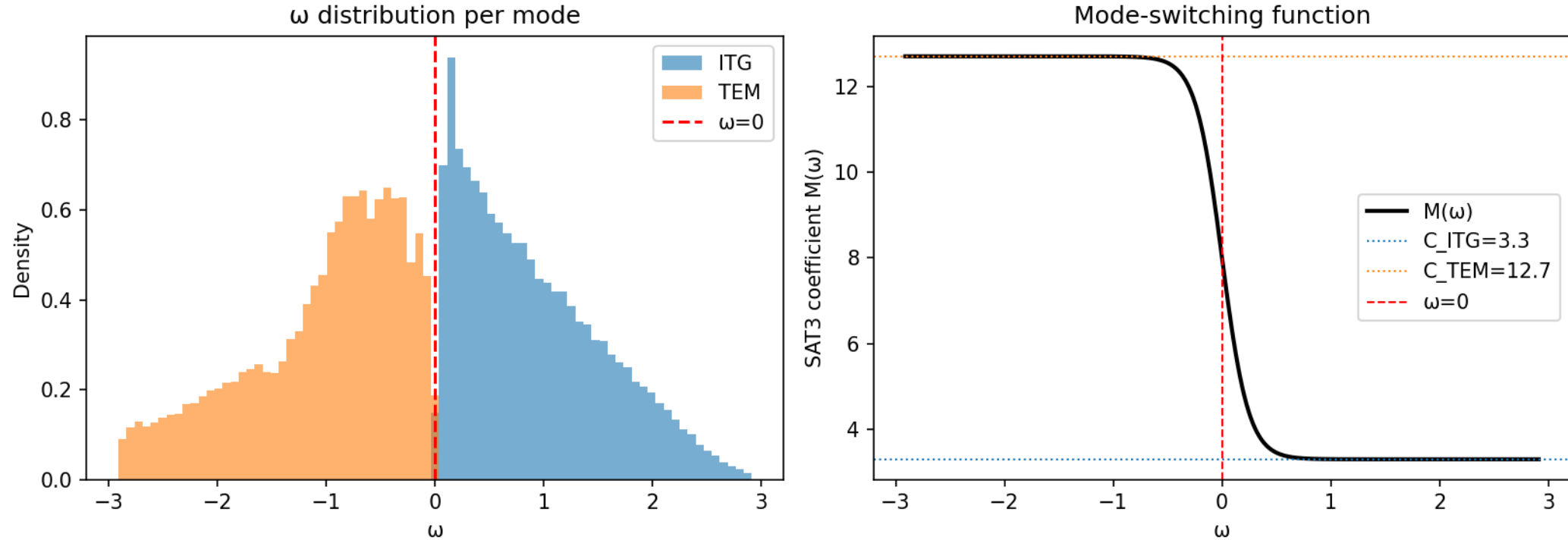
$$\left(\hat{s}, q, \frac{a}{L_{Ti}}, \frac{a}{L_{Te}}, \frac{a}{L_n}, \text{trpeps}, \epsilon \right) \xrightarrow{V3MLP} \left(k_{\max}, \rho_s, \frac{\gamma_{\max} a}{c_s}, R_{ei} \right)$$

Current Performance Plots



Evaluating plots
Trained with GENE Linear NSCC dataset from Youngwoo Cho

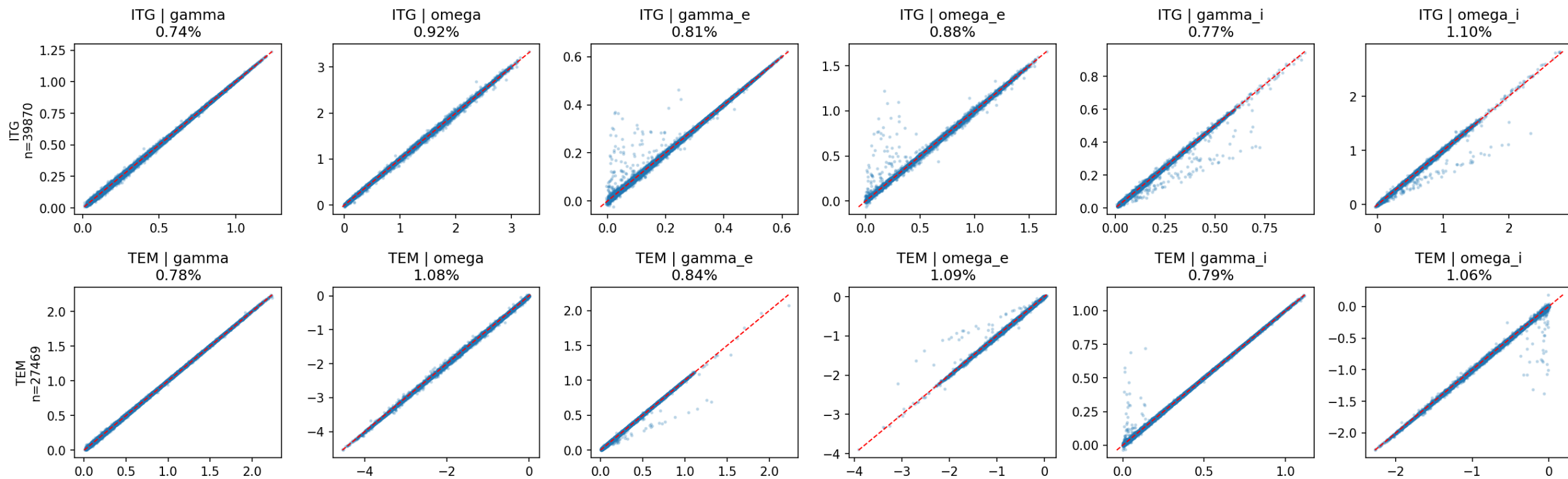
Current Performance Plots



Mode-switch plots
ITG & TEM

Current Performance Plots

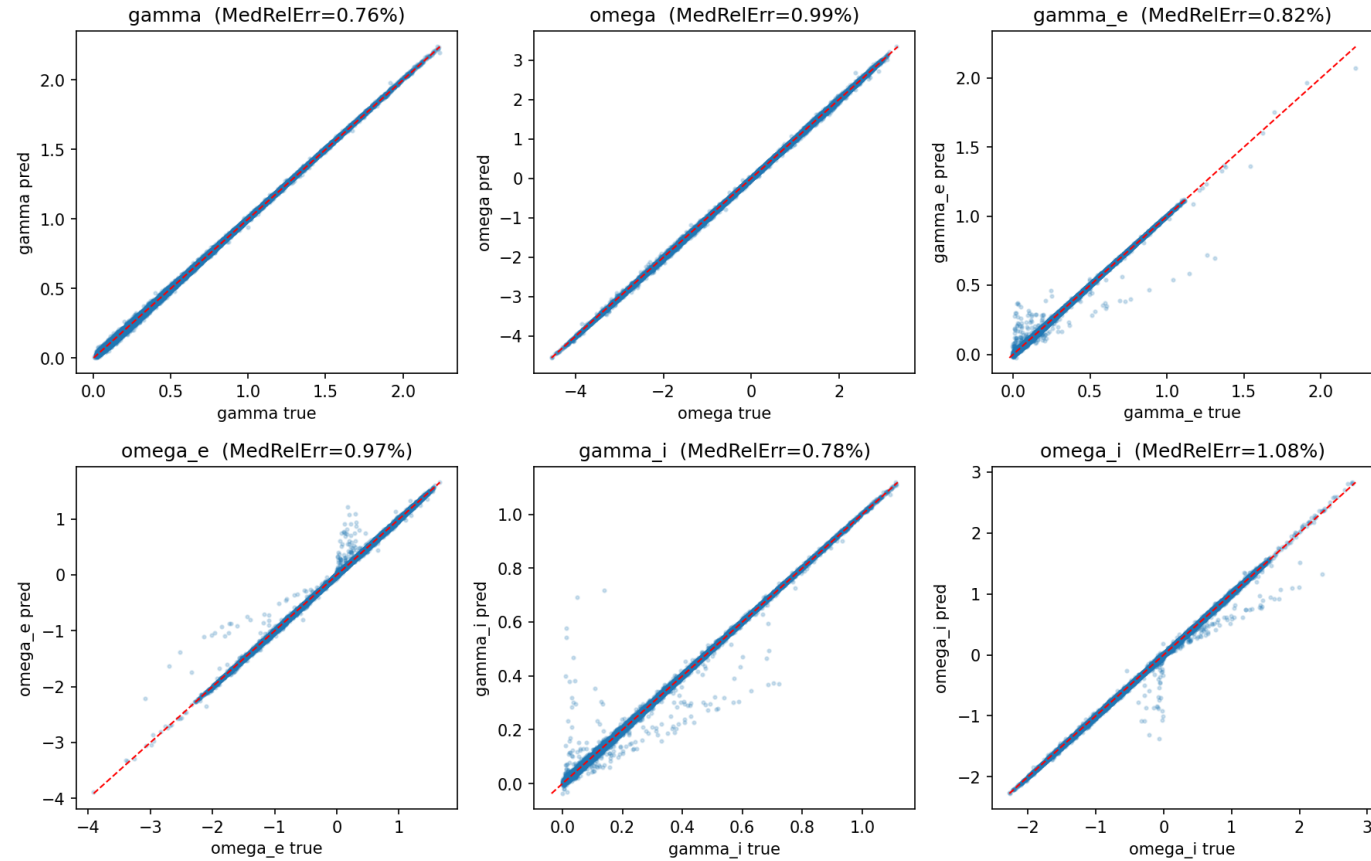
Tatyana V3 — Per-mode parity (GENE linear benchmark)



Mode Benchmark against GENE Linear Dataset
ITG & TEM

Current Performance Plots

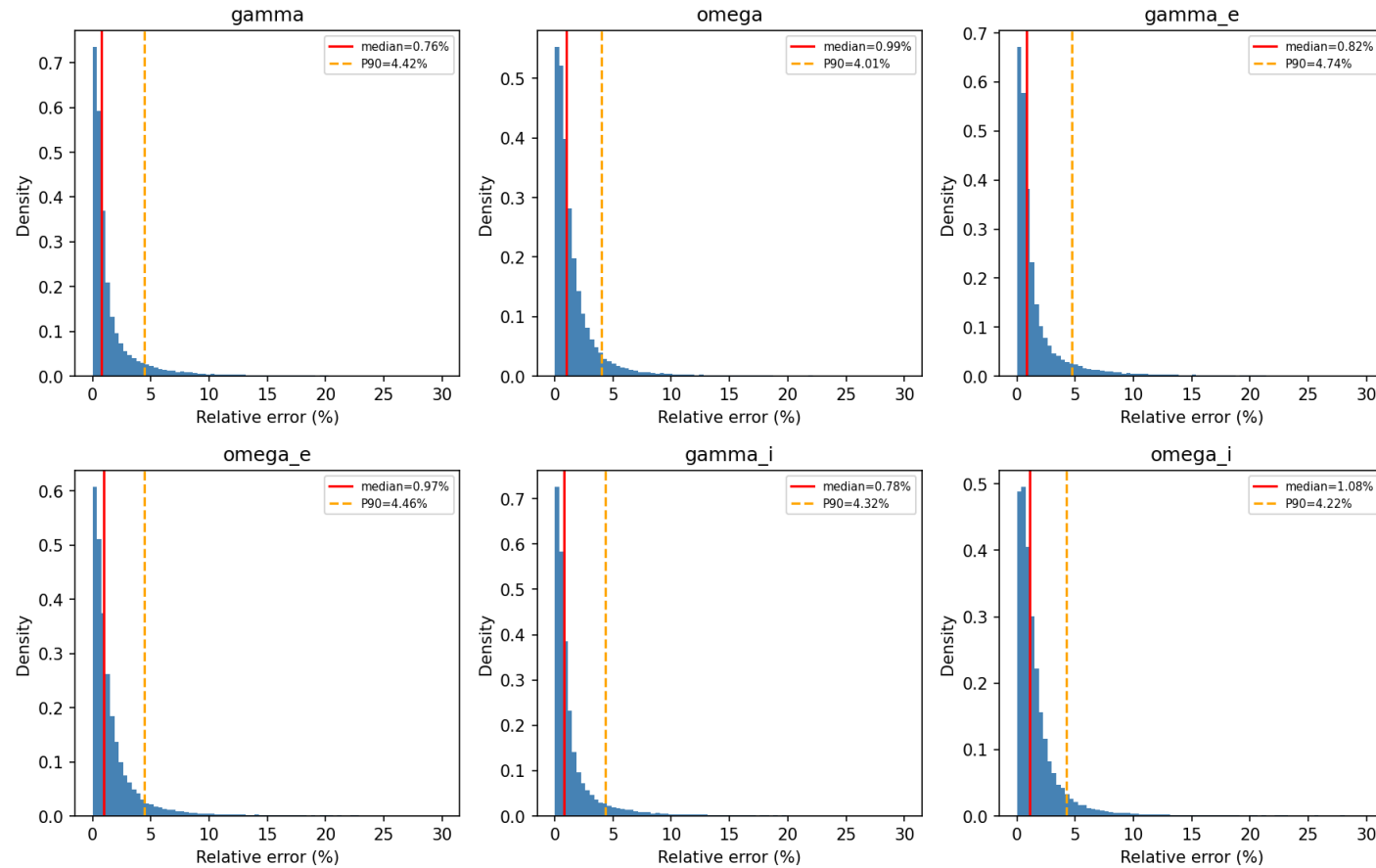
Tatyana V3 — Parity plots (GENE linear benchmark)



Parity Benchmark against GENE Linear Dataset
ITG & TEM

Current Performance Plots

Tatyana V3 — Relative error distributions



Error Distributions

V3 Surrogate prototype available at

<https://github.com/Jekyll-Chan/Neural-Surrogate-TatyanaV3.git>

*This work is inspired in part by: Dudding, H., Casson, F. J., Dickinson, D., Patel, B., Roach, C. M., Belli, E. A., & Staebler, G. (2022)

A new quasilinear saturation rule for tokamak turbulence with application to the isotope scaling of transport.

Nuclear Fusion, 62(9), 096005.

DOI: 10.1088/1741-4326/ac7a4d